

[6402]-2

S.E. (Civil Engineering)

MECHANICS OF STRUCTURE

(2019 Pattern) (Semester - III) (201002)

Time : 2 ½ Hours]

[Max. Marks : 70

Instructions to the candidates :

- 1) Attempt Q.1 or Q.2, Q.3 or Q.4. Q.5. or Q.6 Q.7 or Q.8.
- 2) Figures to the right side indicate full marks.
- 3) Neat diagrams must be drawn wherever necessary.
- 4) Assume Suitable data if necessary.
- 5) Use of scientific calculator is allowed.
- 6) Assessment will be based on complete solution and not on final answer.

- Q1) a) Three wooden planks 200 mm × 20 mm each are connected to form a Symmetrical I section of a beam as shown in Figure 1. A moment of 7 kN.m is applied around the horizontal neutral axis. Find the bending stresses at both extreme fibers of cross section. [8]

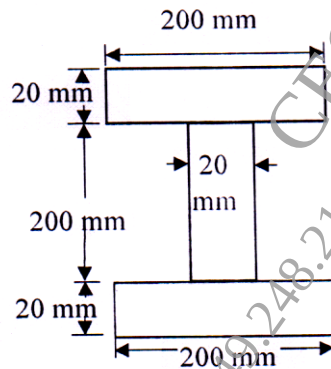


Figure-1

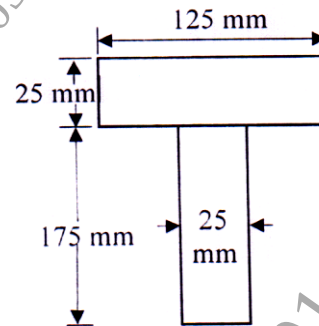


Figure-2

- b) A simply supported beam carries a uniformly distributed load of 30 kN/m over the entire span of 2 m. The cross section of beam is a T section, with its top flange of 125 × 25 mm and web of 175 × 25 mm as shown in Figure 2. Obtain the maximum shear stress and plot a shear stress distribution. [10]

OR

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- Q2) a)** A simply supported beam carries a point load of 100 kN at mid span of the beam of span 2 m. The cross section of beam is a T section, with its top flange of 150×30 mm and web of 30×200 mm as shown in figure 3. Obtain the maximum shear stress and plot a shear stress distribution. [10]
- b)** The cross section of simply supported beam of span 6 m long is shown in figure 4. If permissible stresses are 100 MPa in compression 40 MPa in tension, find safe udl the beam can carry. [8]

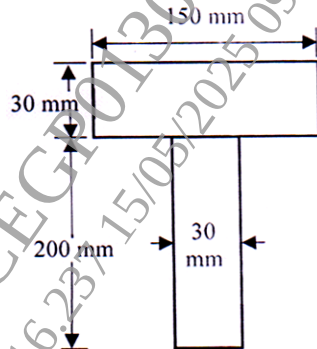


Figure -3

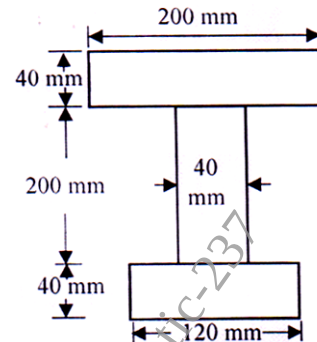


Figure -4

- Q3) a)** Find maximum torque that can be safely applied to a shaft of 100 mm diameter. The permissible angle of twist is 2° in a length of 4 m and shear stress not to exceed 50 MPa. $G = 80 \times 10^3 \text{ N/mm}^2$. [8]
- b)** The plane element in a body is subjected to a tensile normal stresses of 25 N/mm^2 in X direction and shear stress of magnitude 100 N/mm^2 . Calculate the normal, tangential and resultant stresses on a plane inclined at 30° with vertical direction anticlock wise. [9]

OR

- Q4) a)** A hollow shaft with ratio of internal diameter to external diameter $3/5$ is required to transmit 400 kW at 60 rpm with a uniform twisting moment. The shearing stress in the shaft must not exceed 50 N/mm^2 and the twist in a length of 3 m must not exceed 1° . Taking $G = 85 \times 10^3 \text{ N/mm}^2$, determine the minimum external diameter of the shaft satisfying above two conditions. [9]
- b)** The principal tensile stresses at a point are 150 N/mm^2 and 100 N/mm^2 . Find normal and tangential and resultant stress on a plane at 30° with major principal plane. What is angle of obliquity. [8]

- Q5) a)** State the assumptions made in Euler's theory. A 4 m length of tube has buckling load 2 kN when used as a column hinged at both ends. Calculate buckling load for 4.5 m length of the same tube when used as column if,
- (i) Both ends are fixed. (ii) One end fixed and another end is hinged. [9]

- b) A short masonry pillar $600 \text{ mm} \times 600 \text{ mm}$ in section. The pillar carries an eccentric load of 1200 kN . Acting at an eccentricity of 30 mm from the longitudinal axis as shown in figure 5. Find the maximum and minimum stresses on the section column [9]

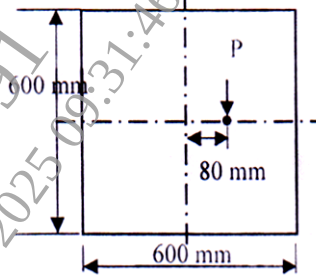


Figure -5

OR

- Q6) a) A hollow C.I. column whose outer diameter is 300 mm has a thickness of 25 mm . It is 4 m long and fixed at both ends. Calculate the safe load by Rankine's formula using a factor of safety of 3. Calculate slenderness ratio and Rankine's critical load. Take $\sigma_c = 500 \text{ N/mm}^2$, $\alpha = \frac{1}{7500}$ and take $E = 2 \times 10^5 \text{ N/mm}^2$. [9]

- b) A column $1.2 \text{ m} \times 1.2 \text{ m}$ is subjected to eccentric load 600 kN as shown in figure 6. Find The stresses at the corner A, B, C and D. Draw stress distribution diagram. [9]

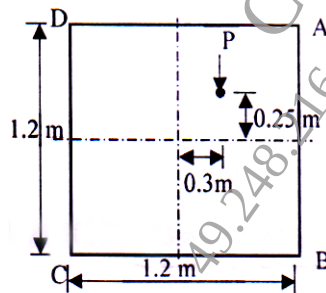


Figure-6

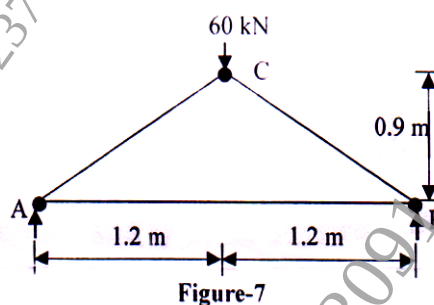


Figure-7

- Q7) a) Calculate the slope and deflection of Simply supported beam of span l carrying a point load at mid span. [8]
- b) Determine the horizontal displacement of the joint C of the pin jointed frame as shown in figure 7. The cross-sectional area of AB is 500 mm^2 and AC and BC is 750 mm^2 . Assume $E = 200 \text{ kN/mm}^2$. [9]

OR

- Q8) a)** A simply supported beam having uniform section is 14 m long and is simply supported at its ends. It carries concentrated load of 12 kN and 8 kN at 3 m and 4.5 m from the two ends respectively as shown in figure 8. I for the beam is $16 \times 10^4 \text{ m}^2$ and $E = 2.1 \times 10^4 \text{ kN/m}^2$. Calculate the deflection of the beam at points under the two loads by Macaulay's method. [9]

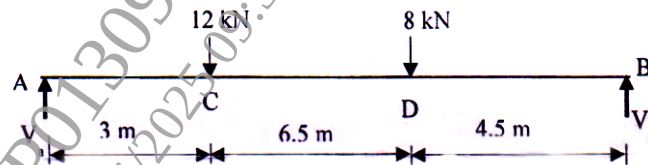


Figure - 8.

- b)** Determine the slope and deflection at free end of cantilever beam of span 'L' meter subjected udl 'w' on entire span. [8]

